

ALGOTHON'26

Problem Statement Documentation

12 Problem Statements • 6 Domains • Unique PS IDs

AI/ML • Web Development • Blockchain • Automation • Data Science • Cybersecurity

Problem Statement Index

Every problem has a unique ID for allocation and tracking

ALG-AI-01

AI / ML | AI Resume & Job Matching System

ALG-WEB-01

Web | Collaborative Project Workspace

ALG-BC-01

Blockchain | Digital Certificate Verification

ALG-AUTO-01

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ALG-DATA-01

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Cybersecurity | Find the Intruder

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AI / ML | Intelligent Document Investigator

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AI / ML

AI Resume & Job Matching System

ALG-AI-01

Recruiters receive hundreds of resumes for a single role. Build a system that analyzes resumes against a job description, ranks them by relevance, and recommends the top candidates.

ALG-AI-01 | AI Resume & Job Matching System

AI / ML • Problem Overview

- Recruiters receive hundreds of resumes for a single role. Build a system that analyzes resumes against a job description, ranks candidates and explains why each candidate matches.

What to Build

Build a web platform where recruiters upload multiple

Must Have

Resume upload
Job-description input
Information extraction
Candidate scoring/ranking
Search and filters
Explainable match result

Innovation / Bonus

Detect unsupported or contradictory claims instead of blindly

ALG-AI-01 | Implementation Guide

AI Resume & Job Matching System • Participant Requirements

- Resume upload
- Job-description input
- Information extraction
- Candidate scoring/ranking
- Search and filters
- Explainable match result

Suggested Tech

Python, NLP, LLMs, embeddings/vector search, OCR, database, React/Next.js

Judging Focus

Accuracy, explainability, useful ranking and robustness to messy resumes.

AI / ML

Intelligent Document Investigator

ALG-AI-02

Information is often scattered across PDFs, images and text documents. Users need answers without manually reading every

ALG-AI-02 | Intelligent Document Investigator

AI / ML • Problem Overview

- Information is often scattered across PDFs, images and text documents. Users need answers without manually reading every document.

What to Build

Build a document investigation platform that accepts

Must Have

Multiple document formats
Extraction/indexing
Natural-language Q&A
Source/section references
Conflict detection
Uncertainty handling

Innovation / Bonus

Identify conflicting documents and communicate uncertainty i

ALG-AI-02 | Implementation Guide

Intelligent Document Investigator • Participant Requirements

- Multiple document formats
- Extraction/indexing
- Natural-language Q&A
- Source/section references
- Conflict detection
- Uncertainty handling

Suggested Tech

RAG, NLP, OCR, embeddings, vector DB, Python, web framework

Judging Focus

Answer quality, source grounding, conflict handling and usable investigation workflow.

Collaborative Project Workspace

ALG-WEB-01

Remote teams often use disconnected tools for tasks, comments, files and progress. Build one collaborative workspace.

ALG-WEB-01 | Collaborative Project Workspace

Web Development • Problem Overview

- Remote teams often use disconnected tools for tasks, comments, files and progress. Build one collaborative workspace.

What to Build

Build a real-time project management workspace where

Must Have

Projects and tasks
Task assignment/status
Comments/files
Deadlines/progress
Dashboard
Real-time updates

Innovation / Bonus

Simultaneous edits to the same task must not silently lose updates

ALG-WEB-01 | Implementation Guide

Collaborative Project Workspace • Participant Requirements

- Projects and tasks
- Task assignment/status
- Comments/files
- Deadlines/progress
- Dashboard
- Real-time updates

Suggested Tech

React/Next.js, WebSockets, Node/FastAPI, PostgreSQL/Supabase

Judging Focus

Real-time reliability, collaboration UX, data consistency and completeness.

Offline-First Application

ALG-WEB-02

Many web applications become unusable when connectivity is unstable. Build an app that remains useful offline and synchronizes data when online.

ALG-WEB-02 | Offline-First Application

Web Development • Problem Overview

- Many web applications become unusable when connectivity is unstable. Build an app that remains useful offline and synchronizes safely later.

What to Build

Build an offline-first web application that caches data,

Must Have

Offline cache
Offline CRUD
Pending changes
Connectivity detection
Synchronization
Sync status

Innovation / Bonus

Conflicting edits from two offline devices must be detected and s

ALG-WEB-02 | Implementation Guide

Offline-First Application • Participant Requirements

- Offline cache
- Offline CRUD
- Pending changes
- Connectivity detection
- Synchronization
- Sync status

Suggested Tech

PWA, service workers, IndexedDB/local storage, React/Next.js, backend APIs

Judging Focus

Offline behavior, synchronization correctness, conflict handling and UX.

BLOCKCHAIN

Digital Certificate Verification

ALG-BC-01

Fake or modified certificates are difficult to verify. Build a blockchain-backed certificate issuing and verification system.

ALG-BC-01 | Digital Certificate Verification

Blockchain • Problem Overview

- Fake or modified certificates are difficult to verify. Build a blockchain-backed certificate issuing and verification system.

What to Build

Build a system where an issuer creates a certificate re

Must Have

Issuer workflow
Unique verification ID
QR/link verification
Public verification page
Status/revocation
Integrity checking

Innovation / Bonus

Verify that certificate data has not changed and that it originates

ALG-BC-01 | Implementation Guide

Digital Certificate Verification • Participant Requirements

- Issuer workflow
- Unique verification ID
- QR/link verification
- Public verification page
- Status/revocation
- Integrity checking

Suggested Tech

Solidity/EVM or equivalent blockchain, hashing, smart contracts, wallet integration

Judging Focus

Actual on-chain verification, integrity, revocation logic and usability.

BLOCKCHAIN

Blockchain Escrow System

ALG-BC-02

Unknown buyers and sellers need a trusted way to transact without relying on a centralized intermediary. Build a blockchain

ALG-BC-02 | Blockchain Escrow System

Blockchain • Problem Overview

- Unknown buyers and sellers need a trusted way to transact without relying on a centralized intermediary. Build a blockchain escrow platform.

What to Build

Build a decentralized escrow platform where a smart

Must Have

Wallet connection
Create escrow
Deposit/lock funds
Seller delivery status
Buyer release
Deadline refund
Basic dispute flow
Transaction history

Innovation / Bonus

Milestone payments, multi-party escrow, reputation or strong

ALG-BC-02 | Implementation Guide

Blockchain Escrow System • Participant Requirements

- Wallet connection
- Create escrow
- Deposit/lock funds
- Seller delivery status
- Buyer release
- Deadline refund
- Basic dispute flow
- Transaction history

Suggested Tech

Solidity, Ethereum-compatible testnet, Hardhat/Foundry, MetaMask, ethers.js/view

Judging Focus

Smart contract security, state transitions, real end-to-end fund flow and edge cases.

AUTOMATION

Visual Workflow Automation

ALG-AUTO-01

People repeat sequences of digital tasks across tools. Build a visual automation platform that lets users define and execute

ALG-AUTO-01 | Visual Workflow Automation

Automation • Problem Overview

- People repeat sequences of digital tasks across tools. Build a visual automation platform that lets users define and execute workflows.

What to Build

Build a visual workflow builder where users drag nodes

Must Have

Visual workflow editor
Triggers
Actions
Conditions
Data transformation
Actual execution
Execution status/logs
Save/load workflows

Innovation / Bonus

Reusable workflows, retries, scheduling, webhooks, parallel br

ALG-AUTO-01 | Implementation Guide

Visual Workflow Automation • Participant Requirements

- Visual workflow editor
- Triggers
- Actions
- Conditions
- Data transformation
- Actual execution
- Execution status/logs
- Save/load workflows

Suggested Tech

React Flow or similar, Node/Python, queues, database, REST APIs

Judging Focus

Real execution engine, correct data passing, reliability and workflow UX.

AUTOMATION

Smart Application Processing

ALG-AUTO-02

Organizations manually process large numbers of applications and documents. Build an automated processing pipeline.

ALG-AUTO-02 | Smart Application Processing

Automation • Problem Overview

- Organizations manually process large numbers of applications and documents. Build an automated processing pipeline.

What to Build

Build a system that accepts applications, validates info

Must Have

Application form
Validation
Missing/inconsistent field detection
Document processing
Categorization
Storage
Notifications

Innovation / Bonus

Incomplete information should be explained and routed for co

ALG-AUTO-02 | Implementation Guide

Smart Application Processing • Participant Requirements

- Application form
- Validation
- Missing/inconsistent field detection
- Document processing
- Categorization
- Storage
- Notifications

Suggested Tech

Workflow automation, document processing, APIs, database, email/notification, o

Judging Focus

Automation completeness, exception handling, accuracy and communication.

DATA SCIENCE

The Mystery Dataset

ALG-DATA-01

Teams receive a large dataset without a predefined business question. They must investigate it and discover meaningful patterns.

ALG-DATA-01 | The Mystery Dataset

Data Science • Problem Overview

- Teams receive a large dataset without a predefined business question. They must investigate it and discover meaningful patterns.

What to Build

Clean and explore the dataset, identify relationships and

Must Have

Data cleaning
Variable understanding
Relationship analysis
Anomaly detection
Visualizations
Hypothesis
Analytical/predictive result

Innovation / Bonus

The strongest submission should explain a meaningful discovery

ALG-DATA-01 | Implementation Guide

The Mystery Dataset • Participant Requirements

- Data cleaning
- Variable understanding
- Relationship analysis
- Anomaly detection
- Visualizations
- Hypothesis
- Analytical/predictive result

Suggested Tech

Python, pandas, NumPy, scikit-learn, visualization libraries

Judging Focus

Quality of analysis, insight depth, visualization and evidence-backed conclusions.

DATA SCIENCE

Predict What Happens Next

ALG-DATA-02

Historical data can reveal patterns that help anticipate future behavior. Build a predictive pipeline from historical data.

ALG-DATA-02 | Predict What Happens Next

Data Science • Problem Overview

- Historical data can reveal patterns that help anticipate future behavior. Build a predictive pipeline from historical data.

What to Build

Explore and clean data, engineer features, select and

Must Have

Exploration/cleaning
Feature engineering
Model selection
Training/validation
Metrics
Unseen predictions

Innovation / Bonus

Use a hidden final evaluation dataset so teams cannot manual

ALG-DATA-02 | Implementation Guide

Predict What Happens Next • Participant Requirements

- Exploration/cleaning
- Feature engineering
- Model selection
- Training/validation
- Metrics
- Unseen predictions

Suggested Tech

Python, pandas, scikit-learn, XGBoost/LightGBM, visualization

Judging Focus

Model quality, validation methodology, feature reasoning and robustness.

CYBERSECURITY

Find the Intruder

ALG-CYBER-01

Thousands of security events can hide an attacker among normal activity. Build a system that identifies suspicious behavior

ALG-CYBER-01 | Find the Intruder

Cybersecurity • Problem Overview

- Thousands of security events can hide an attacker among normal activity. Build a system that identifies suspicious behavior from logs.

What to Build

Analyze authentication, network or server logs to identify suspicious behavior.

Must Have

Log ingestion
Anomaly detection/rules
Suspicious user/IP detection
Event grouping
Incident timeline
Evidence

Innovation / Bonus

Connect multiple events and explain the likely attack sequence.

ALG-CYBER-01 | Implementation Guide

Find the Intruder • Participant Requirements

- Log ingestion
- Anomaly detection/rules
- Suspicious user/IP detection
- Event grouping
- Incident timeline
- Evidence

Suggested Tech

Python, log analysis, anomaly detection, rules/ML, dashboard, database/search

Judging Focus

Detection quality, evidence, correlation across events and explainability.

CYBERSECURITY

Secure the Application

ALG-CYBER-02

Deliberately vulnerable web applications contain weaknesses that attackers can exploit. Identify, demonstrate safely, fix and

ALG-CYBER-02 | Secure the Application

Cybersecurity • Problem Overview

- Deliberately vulnerable web applications contain weaknesses that attackers can exploit. Identify, demonstrate safely, fix and verify them.

What to Build

Inspect the application, identify security weaknesses,

Must Have

Authentication/input inspection
Vulnerability identification
Safe demonstration
Secure fixes
Retesting
Root-cause documentation

Innovation / Bonus

Security fixes must not break legitimate application functionality

ALG-CYBER-02 | Implementation Guide

Secure the Application • Participant Requirements

- Authentication/input inspection
- Vulnerability identification
- Safe demonstration
- Secure fixes
- Retesting
- Root-cause documentation

Suggested Tech

Web security, secure coding, OWASP principles, backend development, testing

Judging Focus

Correct vulnerability identification, safe testing, quality of fixes and regression testing.

Common Submission Expectations

Applies to all 12 problem statements

- Working project with a deployed demo where practical.
- Source-code repository with a clear README.
- Architecture diagram and explanation of major technical decisions.
- Demonstration of the core workflow required by the selected PS.
- Testing evidence and handling of important edge cases.
- Known limitations and future improvements.
- Disclosure of external APIs, datasets and AI-assisted components used.
- Teams may use equivalent technologies unless a PS specifically requires a particular technology.

Evaluation Framework

Common judging criteria

Functionality & Completion

30% • Does the core solution actually work?

Technical Implementation

20% • Quality, architecture and appropriate technology.

Innovation & Problem Understanding

20% • Originality and depth of solution.

User Experience / Presentation

15% • Clarity, usability and demo quality.

Testing, Edge Cases & Reliability

15% • Robustness beyond the happy path.

ALGOTHON'26 PRINCIPLE

**AI MAY HELP YOU BUILD IT.
BUT YOU STILL HAVE TO MAKE IT WORK.**

12 PS • 6 DOMAINS • 1 HACKATHON